JALA Academy
Assignment 9

Abstract Classes in Python

Python solution reference • Code examples are organized by the assignment questions.

How to use this document

Save the code examples into a .py file and run with Python 3. Where a program asks for input, enter values when prompted. For file-handling exercises, create the named text file in the same folder first.

1–6. Abstract and concrete methods

```
from abc import ABC, abstractmethod

class Appliance(ABC):
    @abstractmethod
    def operate(self):
        pass

    def information(self):
        print("This is an appliance.")

class Fan(Appliance):
    def operate(self):
        print("Fan is rotating.")

fan = Fan()
fan.information()  # Inherited non-abstract method
fan.operate()      # Implemented abstract method

# Appliance() would raise TypeError because operate() is abstract.

class Shape(ABC):
    @abstractmethod
    def area(self):
        pass

    def display(self):
        print("Area:", self.area())

class Circle(Shape):
    def __init__(self, radius):
        self.radius = radius
    def area(self):
        return 3.14159 * self.radius ** 2

class Rectangle(Shape):
    def __init__(self, length, width):
        self.length, self.width = length, width
    def area(self):
        return self.length * self.width

Circle(2).display()
Rectangle(4, 5).display()

class IncompleteShape(Shape):
    pass
# IncompleteShape() raises TypeError because area() is not implemented.
```